

Algorithm Use-Case

The baseline algorithm is an implementation of a “peak-normalized asymmetric filter”. It is designed as a pipeline for sensor data conditioning and feature extraction. The provided algorithm applies a linear pre-scaling to normalize the range of the input, uses an asymmetric 4-tap FIR filter to emphasize specific features and suppress noise, and employs a Leaky ReLU activation function to provide non-linear shaping. This serves to make the output signal better suited for neural network classifiers. The overall function is to take some input signal data, normalize it, detect peaks for gain adjustment, and shape the signal for use in classifiers.

How is FPGA Acceleration Beneficial?

FPGA acceleration is beneficial for this algorithm because there is a large potential for parallelism and deterministic latency requirement. We can use the FPGA to create pipelined stages for processing signal samples sequentially in real time. Additionally, the use of an FPGA helps us tackle the fork-join dependency created by the block-based peak detection by using our FIFOs and on-chip BRAM.

Calculated Latency

6831682 cycles * 7.606 ns = 51.962 ms latency

Report Screenshots

RTL	Status	Latency(Clock Cycles)			Interval(Clock Cycles)			Total Execution Time (Clock Cycles)
		min	avg	max	min	avg	max	
VHDL	NA	NA	NA	NA	NA	NA	NA	NA
Verilog	Pass	6831682	6831682	6831682	NA	NA	NA	6831682

Implementation tool: Xilinx Vivado v.2025.1.1
Project: project_1
Solution: hls
Device target: xczu3eg-sbva484-1-e
Report date: Mon Mar 30 19:57:15 EDT 2026

=== Post-Implementation Resource usage ===

SLICE:	0
LUT:	4280
FF:	3658
DSP:	22
BRAM:	352
URAM:	0
LATCH:	0
SRL:	146
CLB:	1078

=== Final timing ===

CP required:	10.000
CP achieved post-synthesis:	6.019
CP achieved post-implementation:	7.606

Timing met